PROPERTY VALUATION PLATFORM

1. Introduction & System Overview

This project is a production-ready Property Valuation Platform frontend built for **Stackly Property Valuation** to replace passive video fatigue with an interactive, studio-grade skill mastery experience. Designed for aspiring engineers, AI practitioners, UX systems designers, and enterprise engineering cohorts, the platform connects a modern multi-page public website with role-based learner and administrator workspaces. Behind the scenes, the architecture combines responsive Flexbox layouts, real-time client-side form validation, interactive competency telemetry, and smooth hardware-accelerated motion effects so every view runs fast and adapts cleanly across desktop, tablet, and mobile screens.

2. Technology Stack & Dependencies

Core Technologies

- **HTML5 & CSS3:** Semantic HTML5 landmarks (`<header>`, `<nav>`, `<main>`, `<section>`, `<aside>`, `<footer>`) paired with modular CSS3 stylesheets inside the dedicated `CSS/` directory, using fluid `clamp()` typography and pure CSS Flexbox layouts across all breakpoints.
- **Vanilla JavaScript (ES6+):** Lightweight, zero-dependency native scripting used to power synchronized active navigation states, filterable course catalogs, interactive 5-axis skill radars, real-time form validation, and Single-Page Application (SPA) dashboard tab switching.
- **GSAP-Compatible Motion & RequestAnimationFrame Engine:** Hardware-accelerated entrance animations, 3D studio card pointer parallax, and smooth numeric counter interpolation with automatic `prefers-reduced-motion` support.
- **AOS (Animate On Scroll):** Lightweight scroll-triggered reveal utility (`[data-aos]`) that activates staggered section entrances as content enters the viewport.

External Libraries & Bundled Assets

- **Font Awesome 6 Free Icons:** Bundled locally via `assets/fontawesome-all.min.css` and `assets/webfonts/fa-solid-900.woff2` to provide crisp vector icons across navigation controls, interactive cards, form alerts, and dashboard sidebars without external CDN blocking.
- **Google Fonts (Outfit & Plus Jakarta Sans):** Imports geometric display weights (`Outfit` 600, 700, 800) for high-impact headlines alongside clean UI body weights (`Plus Jakarta Sans` 400, 500, 600, 700, 800) for optimal readability.
- **Optimized WebP Media Assets:** All logos, hero studio visuals, mentor avatars, and course thumbnails inside `assets/` are encoded in modern `.webp` format (strictly under 100 KB per image) for rapid page loads.

3. File Architecture & Directory Structure

The project is structured as a Multi-Page Application (MPA) for public discovery and authentication, paired with Single-Page Application (SPA) internal dashboards. All HTML views and modular JavaScript controllers reside at the project root, while all stylesheets are organized inside the `CSS/` directory and media resources inside `assets/`:

```
property-valuation/
├── index.html
├── about.html
├── services.html
├── blog.html
├── contact.html
├── login.html
├── signup.html
├── forgot-password.html
├── admin-dashboard.html
├── client-dashboard.html
├── 404.html
├── script.js
├── about.js
├── services.js
├── blog.js
├── contact.js
├── auth.js
├── dashboard.js
├── 404.js
├── README.md
├── CSS/
│   ├── shared.css
│   ├── style.css
│   ├── about.css
│   ├── services.css
│   ├── blog.css
│   ├── contact.css
│   ├── auth.css
│   ├── dashboard.css
│   └── 404.css
└── assets/
    ├── stackly-logo-amber.webp
    ├── hero-valuation-studio.webp
    ├── valuation-residential.webp
    ├── valuation-commercial.webp
    ├── valuation-industrial.webp
    ├── valuation-land.webp
    ├── blog-hero-editorial.webp
    ├── fontawesome-all.min.css
    ├── about/
    ├── avatars/
    └── webfonts/
```

4. Visual Design System (UI/UX)

- **Glassmorphism & Frosted Studio Surfaces:** Crafted using `backdrop-filter: blur(18px) saturate(135%)`, translucent surface fills (`rgba(255, 255, 255, 0.82)`), and subtle indigo border highlights on sticky headers, floating callout notes, and dark advising panels.
- **Pure CSS Flexbox Layout System:** Implements responsive multi-column decks, split hero stages, and metric rows strictly via `display: flex`, `flex-wrap: wrap`, and fluid gap tokens to organize content cleanly across desktop, tablet, and mobile screens.
- **Synchronized Header & Footer Active Navigation:** Both the fixed top navigation bar (`.site-nav`), mobile navigation drawer (`.mobile-nav`), and footer Explore column (`.site-footer__col`) highlight the active page link simultaneously with a tight 2px underline bar positioned directly beneath the link text (`bottom: 0; padding: 2px 0; line-height: 1.15`) with zero unnecessary vertical gap.
- **Interactive Micro-Animations:** Built with smooth CSS transitions (`transform`, `box-shadow`, `border-color`) and subtle hover elevations (`translateY(-3px)`) to provide immediate tactile feedback on buttons, course cards, and tabs.

5. Visual Design Language & Color Palette

The visual identity pairs high-contrast light editorial surfaces with deep indigo-navy studio panels and vibrant royal-indigo accents across all pages and dashboards:

Color Name	CSS Variable	Hex / RGBA Code	Usage Context
Primary Ink	`--ink` / `--nav-ink`	`#0c0e24`	Primary body typography and high-contrast headings on light surfaces
Muted Slate	`--muted`	`#616586`	Secondary descriptions, captions, and supporting lead copy
Pure Surface White	`--surface`	`#ffffff`	Clean card surfaces, header backgrounds, and modal containers
Brand Indigo	`--accent`	`#b45309`	Primary brand accents, progress meters, and interactive highlights
Deep Brand Indigo	`--accent-dark` / `--brand-green`	`#8c3b05`	Active header nav links, underline indicators, and primary CTAs
Bright Studio Violet	`--brand-green-bright`	`#f59e0b` / `#8b7cf8`	Footer active link underline, dark-mode badges, and chart highlights

Color Name	CSS Variable	Hex / RGBA Code	Usage Context
Studio Navy	`--navy`	`#211813`	Primary action buttons, telemetry chart cards, and dark callouts
Footer & Contact Deep	`--footer-navy` / `--contact-bg`	`#16110d` / `#0d071d`	Global footer background and dark-themed Contact/Blog/Services heroes
Dashboard Sidebar Dark	`--dash-dark`	`#170a3f`	Fixed sidebar navigation gradient on Learner and Admin dashboards
Soft Canvas Mint-Grey	`--soft` / `--dash-bg`	`#eef5f3` / `#f3f8f6`	Soft section backgrounds and dashboard workspace canvas
Structural Line	`--line`	`#dce5ee`	Standard card borders, dividers, and input outlines

Semantic State Colors

- **Active / Valid State (Indigo Violet):** `#8b7cf8` / `rgba(124, 102, 245, 0.65)` (Highlights validated inputs, active footer links, and confirmation banners).
- **Error / Invalid State (Coral Red):** `#ff6b6b` / `#cf5d5d` (Highlights invalid form fields with `aria-invalid="true"` and displays field-level error alerts).

Overall Layout

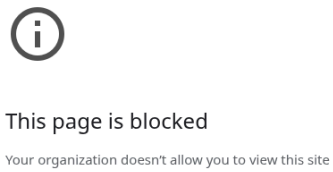


Figure 1: Desktop Hero & Fixed Navigation Layout (index.html)

6. Public-Facing & Internal Architecture Overview

A. Public-Facing Architecture

Home Page (index.html)

- **Hero Section — Interactive Learning Studio:** Features a split Flexbox hero with live counter metrics (`94.8%` completion, `3.2x` faster ramp-up, `4.9★` mentor score) and an interactive 3D telemetry dashboard preview with SVG progress curves and floating callout badges.
- **Interactive Skill Track Workspace & Methodology:** Allows learners to toggle between *AI & Data Science*, *Full-Stack Engineering*, and *Product & UX Systems* tracks with live metric updates, followed by Active Learning Methodology cards.
- **Filterable Flagship Course Catalog:** Provides instant client-side category filtering (`All Programs`, `AI & ML`, `Engineering & Cloud`, `Product Design`) with dynamic card highlighting, ratings, and mentor attribution.
- **5-Axis Competency Radar & Live Mentor Room:** Includes an interactive polygon clip-path radar chart across four role benchmarks and a clickable Live Mentor Coaching Room schedule.

About Page (about.html)

- **Brand Story & Pedagogical Vision:** Shares Stackly's mission to replace passive video lectures with active cloud sandboxes and practitioner mentorship, supported by scroll-triggered animated statistics.
- **Interactive 4-Stage Decision Loop:** Lets visitors inspect the four core learning stages (*01 Diagnostic*, *02 Sandbox*, *03 Critique*, *04 Capstone*) with dynamic center-stage state updates.
- **Expandable Principles Track:** Features an interactive horizontal Flexbox accordion where hovering or focusing each educational principle expands the card and updates the live status bar.

Services / Learning Tracks Page (services.html)

- **Interactive Learning Architecture Map:** Displays a 4-lane interactive service console (*Guided Career Tracks*, *Browser Cloud Labs*, *1:1 Mentor Studios*, *Enterprise Academies*) with real-time status summaries.
- **Offer Deck & 4-Step Delivery Path:** Showcases equal-height Flexbox offer cards and a 4-stage delivery rail (*Diagnose*, *Build*, *Verify*, *Deploy*) with interactive outcome telemetry.

Blog Page (blog.html)

- **Editorial Learning Guides & Search:** Presents engineering deep dives, AI retrieval benchmarks, and career roadmaps with a search bar and topic filter pills.
- **Interactive Field Tabs:** Implements full keyboard arrow-key navigation (`ArrowRight`, `ArrowLeft`, `ArrowDown`, `ArrowUp`) across three technical methodology panels (*Architecture framing*, *Sandbox execution*, *Portfolio proof*).

Contact Page (contact.html)

- **Admissions & Advising Inquiry Form:** Provides structured fields for Full Name, Work or Student Email, Learning Focus topic selection, and Background/Goal message inside a frosted dark indigo panel.
- **Comprehensive Client-Side Form Validation:** Validates each field on `blur`, `input`/`change`, and `submit` — checking name length and character format (`/^[A-Za-z][A-Za-z\s.'-]{1,79}$/`), strict email syntax, required topic selection, and minimum 10-character message length. Displays targeted per-field error messages (`.contact-field__error`), sets `aria-invalid="true"` red border highlights, focuses the first invalid control, and reveals a confirmation banner upon valid submission.
- **Starting Pathways & Admissions FAQ Accordion:** Guides visitors through three entry doors and a 4-item interactive FAQ accordion with `aria-expanded` state management.

Authentication Portals (login.html, signup.html & forgot-password.html)

- **Split-Screen Studio Auth UI:** Pairs an ambient indigo visual panel with a clean form container featuring role selection (*Admin* vs. *Client/Learner*), password visibility toggle buttons (Font Awesome eye/eye-slash icons), and a 4-bar real-time password strength meter on Sign Up.
- **Client-Side Security & Role Routing:** Enforces email regex patterns, role selection, and password complexity rules before dynamically routing authenticated users to `admin-dashboard.html` or `client-dashboard.html`.

Error Page (404.html)

- **Fixed, Non-Scrolling Viewport:** Locks `html, body` to `position: fixed; inset: 0; height: 100dvh; overflow: hidden` with all CSS keyframe animations and transitions disabled (`animation: none; transition: none`).
- **Centered 404 NOT FOUND Branding & Dual Navigation:** Displays a centered card with bold **404 NOT FOUND** typography and two action buttons directly below — **Go to Home** (navigating to `index.html`) and **Previous Page** (invoking `window.history.back()` with automatic fallback to `index.html`).

B. Internal Architecture (Dashboards)

Both dashboards operate on a fast Single-Page Application (SPA) tab-switching architecture powered by `dashboard.js` and styled by `CSS/dashboard.css`. The navigation sidebar (`.dashboard-sidebar`) is configured with `position: fixed; top: 0; left: 0; bottom: 0; height: 100dvh; overflow-y: auto` so it remains permanently fixed on desktop viewports while the main workspace shell (`.dashboard-shell`) scrolls independently with a matching left margin offset (272px on desktop, 248px below 1080px, and a slide-out off-canvas drawer below 900px).

Learner Dashboard (client-dashboard.html)

- **Fixed Sidebar Navigation & Learner Identity:** Provides instant tab switching across *Overview*, *Skill Radar*, *My Labs*, *Mentor Notes*, and *Preferences*, while hydrating the topbar avatar initial and email from the URL query string.
- **Interactive Learner Panels:** Displays active sprint progress (`94.2` skill score, `28 / 30` cloud labs passed), 5-axis competency meters, verified sandbox repository logs, mentor code-review notes, and IDE preferences.

Admin Dashboard (admin-dashboard.html)

- **Fixed Sidebar Navigation & Academy Controls:** Connects operational views across *Overview*, *Learners*, *Telemetry*, *Courses*, and *Settings* with a persistent fixed navigation rail and Log Out action.
- **Cohort & Infrastructure Management Panels:** Tracks academy-wide KPIs (`1,480` active learners, 24 published tracks, `180` active cloud sandboxes, 96% completion rate), pending cohort applications, mentor critique SLAs, and curriculum release queues.

7. Animation & Performance Standards

- **Hardware-Accelerated Motion:** Restricts interactive transitions and entrance reveals on content pages to compositor-friendly properties (`transform` and `opacity`) for smooth 60fps rendering across desktop and mobile devices.
- **Strict Reduced-Motion & Fixed 404 Compliance:** Honors `@media (prefers-reduced-motion: reduce)` globally across all scripts and stylesheets, and completely removes animations and scrolling on `404.html` for instant, distraction-free recovery.
- **Non-Blocking UI Rendering:** Uses lightweight inline SVG charts, CSS clip-path polygons, and local WOFF2 icon fonts to eliminate third-party charting overhead and layout shifts.

8. Authentication & Error Handling

Client-Side Security & Validation

- **Multi-Form Validation Rules:** Enforces field-level and form-level validation across Contact (`contact.js`), Login/Sign Up/Password Recovery (`auth.js`), and Footer Newsletter forms (`script.js`), checking name patterns, email syntax, topic/role dropdown selections, and password/message length requirements.
- **Real-Time Error Guidance & Accessibility:** Marks invalid inputs with `aria-invalid="true"` and `.is-invalid` border highlights, populates linked `aria-describedby` alert containers in real time, and moves keyboard focus to the first invalid field on submission.
- **URL-Driven Role Routing:** Evaluates the selected workspace role (`admin` or `client`) upon login validation and routes directly to `admin-dashboard.html?email=...` or `client-dashboard.html?email=...` without relying on persistent browser storage.

9. Responsive & Performance Methodologies

- **Fluid Flexbox Layouts & Adaptive Breakpoints:** Uses fluid `clamp()` spacing and media queries at `1120px`, `1080px`, `980px`, `900px`, `820px`, `760px`, `600px`, `480px`, and `360px` so every section adapts cleanly from widescreen monitors down to compact smartphones.
- **Responsive Mobile Drawers:** Converts the desktop header navigation into an accessible full-screen slide-in drawer below `760px` and transforms the fixed dashboard sidebar into an off-canvas overlay menu below `900px` with Escape-key and backdrop-click dismissal.
- **Optimized WebP Asset Delivery:** Uses compressed `.webp` imagery across all hero banners, course cards, and avatars to keep total archive weight compact and responsive.

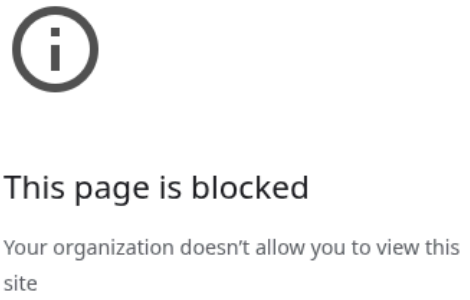


Figure 2: Mobile Responsive Header & Flagship Course Catalog View (390px Viewport)

10. SEO & Asset Optimization

- **Semantic HTML5 Markup & Metadata:** Every page includes explicit `<title>` tags, descriptive `<meta name="description">` attributes, responsive viewport declarations, and hierarchical heading structure (`<h1>`–`<h3>`) with ARIA labels and `aria-current="page"` states.
- **Self-Contained Local Assets:** All stylesheets in `CSS/`, scripts at the project root, WebP images in `assets/`, and Font Awesome WOFF2 fonts in `assets/webfonts/` resolve locally with zero

broken links.

11. Future Scalability Insights

While the current static MPA and SPA frontend delivers a fast, responsive user experience, scaling to a full production academy environment supports the following backend integrations:

- **Backend Framework & API Integration:** Migrating the modular HTML/JS views into a component framework (such as React, Next.js, or Angular) backed by Node.js or .NET REST/GraphQL APIs will enable server-side authentication, persistent cohort state management, and dynamic course authoring.
- **Relational Database & Vector Storage:** Connecting learner profiles, lab submission logs, and Contact roadmap inquiries to PostgreSQL alongside a vector database will secure user records and power AI-driven curriculum diagnostics and semantic course search.
- **WebSocket & Container Orchestration Integration:** Pairing the learner and admin dashboards with persistent WebSocket streams and Kubernetes sandbox orchestration will power real-time browser IDE container provisioning, live unit-test streaming, and instant mentor studio notifications.